

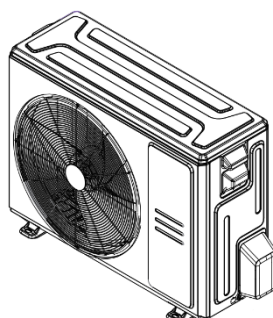
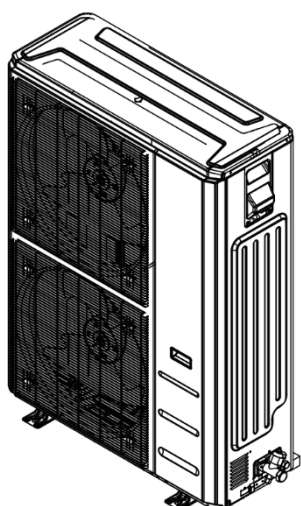
PRODUCT SPECIFICATION

Z-SERIES – HEAT PUMP, 1.5-5 TON
MODEL NO. SPHP-Z-VO##BB



CONTENTS:

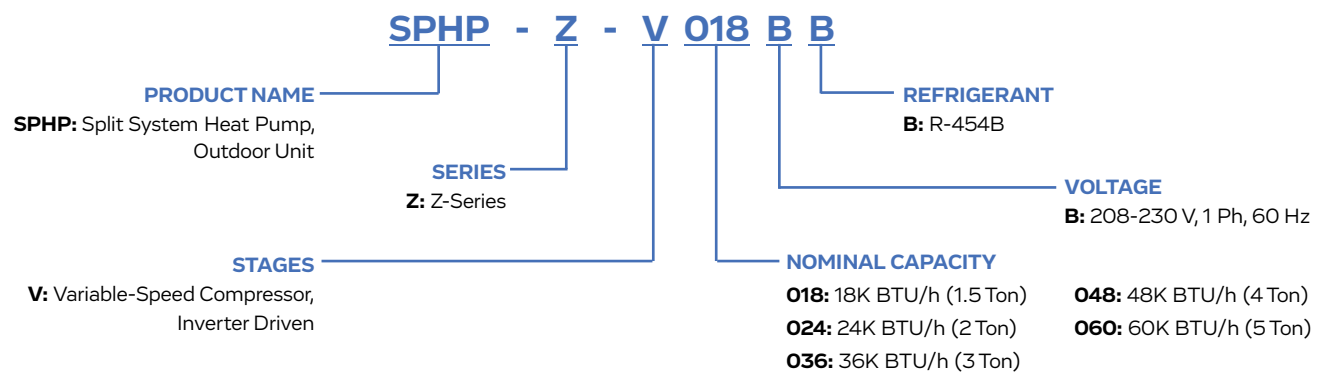
1. Features.....	1
2. Nomenclature.....	2
3. Specifications.....	2
4. Dimensions and Weight.....	3
5. Expanded Ratings.....	4



1. FEATURES

- R-454B (A2L) refrigerant. Low flammability, low toxicity, and low GWP.
- High efficiency, up to 19 SEER2 and 10.1 HSPF2
- Energy-efficient, variable-speed, inverter-driven compressor.
- Internal protection against high temperature motor overload conditions.
- Equipped with chassis heater and crankcase heater as standard.
- Fin and tube condenser with hydrophilic aluminum fins.
- Flared refrigerant line connections.
- Powder-coated galvanized steel construction.
- Troubleshooting LED display.
- 2-way wired communication or standard 24 VAC controls.

2. NOMENCLATURE



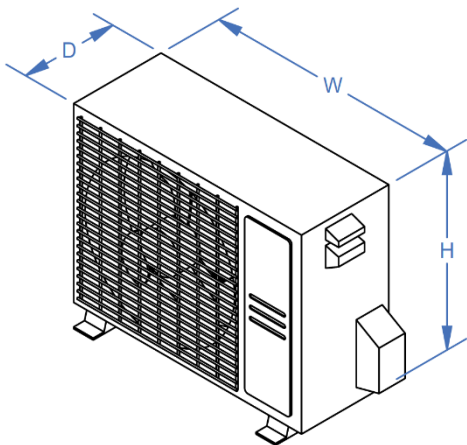
3. SPECIFICATIONS

ITEM	UNITS	SPHP-Z-V018BB	SPHP-Z-V024BB	SPHP-Z-V036BB	SPHP-Z-V048BB	SPHP-Z-V060BB
AC Pro #	-	84231	84232	84233	84234	84235
GENERAL						
System Type	-	Split, Central Ducted				
Unit Type	-	Outdoor Unit, Heat Pump				
Matched Indoor Unit	-	AIRH-Z-V018V06C	AIRH-Z-V024V08C	AIRH-Z-V036V12C	AIRH-Z-V048V16C	AIRH-Z-V060V18C
Controls Type*1	-	2-Way Communication W/ Matched Indoor Unit, 24V Controls (2H/2C)				
Nominal Capacity	BTU/h	18K	24K	36K	48K	60K
PERFORMANCE*2						
Rated Cooling Capacity*3	BTU/h	18,000	24,000	36,000	48,000	54,000
EER2*3	-	12.5	12	12	12	12
SEER2*3	-	19	18.6	17.7	17.5	17.5
Rated Heating Capacity*4	BTU/h	18,000	24,000	36,000	48,000	55,000
HSPF2*4	-	10.1	10	10	9.5	9.5
ELECTRICAL						
Input Voltage, Phases, Frequency*5	V, Ph, Hz	208-230, 1, 60				
Minimum Circuit Ampacity (MCA)	A	16	19	26	38	40
Maximum Overcurrent Protection (MOCP)	A	20	20	30	40	40
COMPRESSOR						
Type	-	Rotary, Inverter, DC, Variable				
Rated Load Amps	A	10.5	14	18	28	31
OUTDOOR FAN(S)						
Motor Type	-	Brushless DC, Variable				
Airflow	CFM	1766 (Max)	1766 (Max)	3002 (Max)	3037 (Max)	2649 (Total, Max)
Full Load Amps	A	0.9	1	3	2	2X: 1.2
REFRIGERATION SYSTEM						
Refrigerant	-	R-454B				
Refrigerant Pre-Charge	lb – oz	4 – 10	4 – 10	7 – 15	9 – 13	11 – 7.5
Pre-Charged for Lineset Length	ft	24.5				
Additional Charge Per ft	oz/ft	0.7				
Max Lineset Length	ft	164		213	246	
Max Height Difference	ft	82		98.5		
Connection Method	-	Flared				
Liquid Connection Size (OD)	in	3/8				
Vapor Connection Size (OD)	in	3/4				
Metering Device	-	EEV				
OPERATION RANGE						
Outdoor Temperature (Cooling)	°F	-22 – 122				
Outdoor Temperature (Heating)	°F	-22 – 75				
NOISE LEVEL						
Sound Pressure Level	dB(A)	56.5	59	62.5	65	64.5

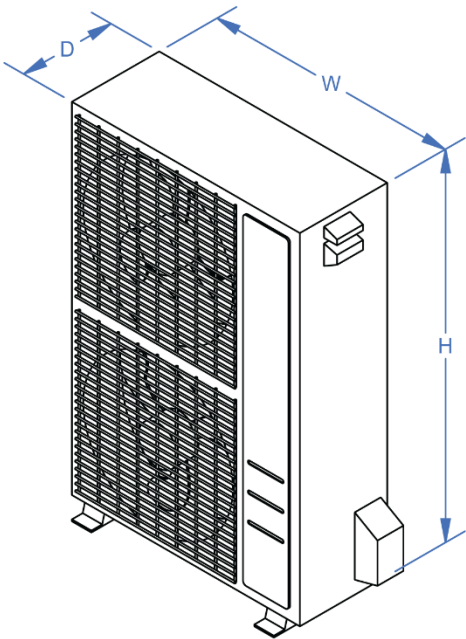
NOTES:
*1. 2-Way communication is required for full capability. Performance is reduced with 24V controls. Refer to *Installation Instructions*. *2. Based on matched system rating. Refer to AHRI directory. * Where indicated, specifications apply at the following conditions and temperatures: *3. Indoor DB 80°F/WB 67°F, Outdoor DB 95°F/WB 75°F; *4. Indoor DB 70°F/WB 60°F, Outdoor DB 47°F/WB 43°F. *5. Operating voltage is 187-253 VAC.

4. DIMENSIONS AND WEIGHT

NOMINAL CAPACITY:
18K, 24K, 36K, and 48K



NOMINAL CAPACITY:
60K



NOTE: Images for reference only. Actual appearance may vary.

ITEM	UNITS	SPHP-Z-V018BB	SPHP-Z-V024BB	SPHP-Z-V036BB	SPHP-Z-V048BB	SPHP-Z-V060BB
Height (H)	in	26-1/2	26-1/2	31-7/8	38-3/8	52-1/2
Width (W)	in	35-1/16	35-1/16	37-1/4	38-9/16	37-1/2
Depth (D)	in	13-7/16	13-7/16	16-1/8	16-5/16	16-5/16
Weight	in	101.5	102.5	167	201.5	243

5. EXPANDED RATINGS

OUTDOOR UNIT:MODEL NO. SPHP-Z-V018BBAC PRO NO. 84231

INDOOR UNIT:MODEL NO. AIRH-Z-V018V06CAC PRO NO. 84226

COOLING MODE

Indoor Airflow (CFM)		530									620								
Outdoor DB (°F)	Indoor WB (°F)	60.8			64.4			67			60.8			64.4			67		
	Indoor DB (°F)	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80
77	T.C.	18.94	18.97	18.99	20.08	20.11	20.13	21.28	21.31	21.33	20.40	20.43	20.45	21.63	21.66	21.68	22.92	22.95	22.98
	S.C.	9.16	10.76	12.24	10.29	12.09	13.75	11.56	13.58	15.45	10.62	12.48	14.20	11.94	14.02	15.95	13.41	15.76	17.92
	I.P.	0.94	0.94	0.94	0.99	1.00	1.00	1.05	1.06	1.06	1.01	1.01	1.01	1.07	1.07	1.07	1.14	1.14	1.14
82	T.C.	19.30	19.33	19.35	20.46	20.49	20.51	21.69	21.71	21.74	20.79	20.82	20.84	22.04	22.07	22.09	23.36	23.39	23.41
	S.C.	9.45	11.10	12.63	10.62	12.47	14.19	11.93	14.01	15.94	10.96	12.88	14.65	12.32	14.47	16.46	13.84	16.26	18.50
	I.P.	1.05	1.05	1.05	1.11	1.11	1.11	1.18	1.18	1.18	1.13	1.13	1.13	1.20	1.20	1.20	1.27	1.27	1.27
86	T.C.	19.23	19.25	19.27	20.38	20.41	20.43	21.60	21.63	21.65	20.71	20.74	20.76	21.96	21.98	22.00	23.27	23.30	23.32
	S.C.	9.53	11.20	12.74	10.71	12.58	14.31	12.03	14.14	16.08	11.06	12.99	14.78	12.43	14.60	16.61	13.96	16.40	18.66
	I.P.	1.24	1.24	1.24	1.31	1.32	1.32	1.39	1.39	1.40	1.33	1.34	1.34	1.42	1.42	1.42	1.50	1.50	1.50
95	T.C.	19.06	19.08	19.10	20.21	20.23	20.25	21.41	21.44	21.46	20.53	20.55	20.58	21.76	21.79	21.81	23.06	23.09	23.12
	S.C.	9.45	11.10	12.63	10.62	12.47	14.19	11.93	14.01	15.94	10.96	12.88	14.65	12.32	14.47	16.46	13.84	16.26	18.50
	I.P.	1.67	1.67	1.67	1.77	1.77	1.77	1.87	1.88	1.88	1.80	1.80	1.80	1.90	1.91	1.91	2.02	2.02	2.02
104	T.C.	16.38	16.40	16.42	17.36	17.38	17.40	18.40	18.43	18.45	17.64	17.66	17.68	18.70	18.72	18.74	19.82	19.85	19.87
	S.C.	8.32	9.78	11.12	9.35	10.99	12.50	10.51	12.34	14.04	9.65	11.34	12.90	10.85	12.75	14.50	12.19	14.32	16.29
	I.P.	1.63	1.63	1.63	1.72	1.73	1.73	1.83	1.83	1.83	1.75	1.75	1.76	1.86	1.86	1.86	1.97	1.97	1.97
110	T.C.	14.77	14.79	14.80	15.66	15.68	15.69	16.59	16.62	16.63	15.91	15.93	15.95	16.87	16.89	16.90	17.87	17.90	17.92
	S.C.	7.78	9.14	10.40	8.74	10.27	11.68	9.82	11.54	13.13	9.03	10.60	12.06	10.14	11.92	13.55	11.39	13.39	15.23
	I.P.	1.60	1.60	1.60	1.70	1.70	1.70	1.80	1.80	1.80	1.72	1.73	1.73	1.83	1.83	1.83	1.94	1.94	1.94
115	T.C.	13.46	13.48	13.49	14.27	14.29	14.30	15.12	15.14	15.16	14.50	14.52	14.53	15.37	15.39	15.41	16.29	16.31	16.33
	S.C.	7.17	8.43	9.59	8.06	9.47	10.77	9.06	10.64	12.11	8.32	9.78	11.12	9.35	10.99	12.50	10.51	12.34	14.04
	I.P.	1.58	1.59	1.59	1.68	1.68	1.68	1.78	1.78	1.78	1.71	1.71	1.71	1.81	1.81	1.81	1.92	1.92	1.92
122	T.C.	11.72	11.73	11.75	12.42	12.44	12.45	13.17	13.18	13.20	12.62	12.64	12.65	13.38	13.40	13.41	14.18	14.20	14.21
	S.C.	6.39	7.51	8.54	7.18	8.44	9.60	8.07	9.48	10.78	7.41	8.71	9.91	8.33	9.79	11.13	9.36	11.00	12.51
	I.P.	1.56	1.56	1.56	1.65	1.66	1.66	1.75	1.75	1.76	1.68	1.68	1.68	1.78	1.78	1.78	1.89	1.89	1.89

T.C. = Total Cooling Capacity (BTU/h x 1000)I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.

S.C. = Sensible Cooling Capacity (BTU/h x 1000)Highlight = Tested Data

HEATING MODE

Indoor Airflow (CFM)			530					620				
Outdoor DB (°F)	Outdoor WB (°F)	Indoor DB (°F)	60.8	68	70	71.6	75.2	60.8	68	70	71.6	75.2
5	3	T.C.	18.34	17.97	17.88	17.40	17.24	19.75	19.36	19.26	19.18	19.01
		I.P.	2.55	2.57	2.57	2.58	2.59	2.60	2.62	2.63	2.63	2.64
17	15	T.C.	18.66	18.29	18.20	17.71	17.55	20.10	19.70	19.60	19.52	19.35
		I.P.	2.33	2.35	2.36	2.36	2.37	2.38	2.40	2.41	2.41	2.42
35	33	T.C.	18.70	18.33	18.23	17.74	17.58	20.14	19.74	19.64	19.56	19.39
		I.P.	1.93	1.95	1.95	1.96	1.96	1.97	1.99	1.99	1.99	2.00
47	43	T.C.	22.01	21.57	21.46	20.88	20.69	23.71	23.23	23.12	23.02	22.82
		I.P.	1.84	1.86	1.86	1.86	1.87	1.88	1.89	1.90	1.90	1.91
62	56.5	T.C.	15.94	15.62	15.55	15.12	14.99	17.17	16.83	16.74	16.68	16.53
		I.P.	1.40	1.41	1.42	1.42	1.43	1.43	1.44	1.45	1.45	1.45

T.C. = Total Heating Capacity (BTU/h x 1000)I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.

Highlight = Tested Data

5. EXPANDED RATINGS (CONTINUED)

OUTDOOR UNIT: MODEL NO. SPHP-Z-V024BB AC PRO NO. 84232

INDOOR UNIT: MODEL NO. AIRH-Z-V024V08C AC PRO NO. 84227

COOLING MODE

Indoor Airflow (CFM)		694									824								
Outdoor DB (°F)	Indoor WB (°F)	60.8			64.4			67			60.8			64.4			67		
	Indoor DB (°F)	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80
77	T.C.	17.97	17.99	18.01	19.05	19.07	19.09	20.19	20.21	20.24	19.95	19.97	19.99	21.15	21.17	21.19	22.41	22.44	22.47
	S.C.	8.43	9.90	11.26	9.47	11.12	12.65	10.64	12.50	14.22	10.39	12.20	13.88	11.67	13.71	15.60	13.11	15.40	17.52
	I.P.	0.95	0.95	0.95	1.01	1.01	1.01	1.07	1.07	1.07	1.05	1.05	1.06	1.12	1.12	1.12	1.18	1.18	1.19
82	T.C.	18.70	18.72	18.74	19.83	19.85	19.87	21.01	21.04	21.06	20.76	20.79	20.81	22.01	22.04	22.06	23.32	23.36	23.38
	S.C.	8.88	10.43	11.87	9.98	11.73	13.34	11.21	13.17	14.99	10.95	12.86	14.63	12.30	14.45	16.44	13.82	16.24	18.47
	I.P.	1.14	1.14	1.14	1.20	1.21	1.21	1.28	1.28	1.28	1.26	1.26	1.26	1.34	1.34	1.34	1.42	1.42	1.42
86	T.C.	20.00	20.02	20.04	21.20	21.23	21.25	22.47	22.50	22.52	22.20	22.23	22.25	23.54	23.56	23.59	24.94	24.98	25.00
	S.C.	9.62	11.30	12.85	10.81	12.70	14.44	12.14	14.27	16.23	11.85	13.93	15.84	13.32	15.65	17.80	14.97	17.58	20.00
	I.P.	1.49	1.49	1.49	1.58	1.58	1.58	1.67	1.67	1.67	1.65	1.65	1.65	1.75	1.75	1.75	1.85	1.86	1.86
95	T.C.	22.91	22.95	22.97	24.30	24.32	24.35	25.75	25.78	25.81	25.44	25.47	25.50	26.97	27.01	27.03	28.58	28.62	28.65
	S.C.	11.02	12.95	14.73	12.39	14.55	16.55	13.92	16.35	18.60	13.59	15.96	18.16	15.27	17.94	20.40	17.15	20.15	22.92
	I.P.	2.28	2.28	2.28	2.41	2.42	2.42	2.56	2.56	2.56	2.53	2.53	2.53	2.68	2.68	2.68	2.84	2.84	2.85
104	T.C.	20.46	20.49	20.51	21.70	21.72	21.74	22.99	23.02	23.05	22.72	22.75	22.77	24.09	24.12	24.14	25.53	25.56	25.59
	S.C.	10.09	11.85	13.48	11.34	13.32	15.15	12.74	14.96	17.02	12.44	14.61	16.62	13.98	16.42	18.68	15.70	18.44	20.98
	I.P.	2.26	2.26	2.26	2.40	2.40	2.40	2.54	2.54	2.54	2.51	2.51	2.51	2.66	2.66	2.66	2.82	2.82	2.82
110	T.C.	18.99	19.02	19.04	20.14	20.16	20.18	21.34	21.37	21.39	21.09	21.12	21.14	22.36	22.38	22.41	23.69	23.72	23.75
	S.C.	9.71	11.40	12.97	10.91	12.82	14.58	12.25	14.40	16.38	11.96	14.06	15.99	13.45	15.80	17.97	15.10	17.75	20.19
	I.P.	2.25	2.25	2.25	2.38	2.39	2.39	2.53	2.53	2.53	2.50	2.50	2.50	2.65	2.65	2.65	2.81	2.81	2.81
115	T.C.	16.93	16.95	16.97	17.95	17.97	17.99	19.02	19.05	19.07	18.79	18.82	18.84	19.93	19.95	19.97	21.12	21.14	21.17
	S.C.	8.75	10.28	11.70	9.84	11.56	13.15	11.05	12.98	14.77	10.79	12.68	14.42	12.13	14.24	16.20	13.62	16.00	18.20
	I.P.	2.05	2.06	2.06	2.18	2.18	2.18	2.31	2.31	2.31	2.28	2.28	2.29	2.42	2.42	2.42	2.56	2.57	2.57
122	T.C.	14.18	14.19	14.21	15.03	15.05	15.06	15.93	15.95	15.97	15.74	15.76	15.78	16.69	16.71	16.72	17.68	17.71	17.73
	S.C.	7.50	8.81	10.02	8.43	9.90	11.26	9.47	11.12	12.66	9.24	10.86	12.36	10.39	12.20	13.88	11.67	13.71	15.60
	I.P.	1.79	1.80	1.80	1.90	1.90	1.91	2.02	2.02	2.02	1.99	2.00	2.00	2.11	2.11	2.12	2.24	2.24	2.24

T.C. = Total Cooling Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.

S.C. = Sensible Cooling Capacity (BTU/h x 1000) Highlight = Tested Data

HEATING MODE

Indoor Airflow (CFM)			694					824				
Outdoor DB (°F)	Outdoor WB (°F)	Indoor DB (°F)	60.8	68	70	71.6	75.2	60.8	68	70	71.6	75.2
5	3	T.C.	19.82	19.43	19.33	19.11	18.94	22.01	21.57	21.46	21.37	21.18
		I.P.	2.88	2.90	2.91	2.91	2.92	2.93	2.96	2.96	2.97	2.98
17	15	T.C.	25.00	24.50	24.38	24.11	23.89	27.76	27.20	27.07	26.96	26.72
		I.P.	1.21	1.22	1.22	1.22	1.23	1.23	1.24	1.25	1.25	1.25
35	33	T.C.	23.05	22.59	22.47	22.22	22.02	25.59	25.07	24.95	24.85	24.63
		I.P.	2.44	2.46	2.47	2.47	2.48	2.49	2.51	2.52	2.52	2.53
47	43	T.C.	26.85	26.31	26.18	25.88	25.65	29.81	29.21	29.06	28.95	28.69
		I.P.	2.37	2.39	2.40	2.40	2.41	2.42	2.44	2.45	2.45	2.46
62	56.5	T.C.	22.03	21.59	21.48	21.24	21.05	24.46	23.97	23.85	23.75	23.54
		I.P.	2.03	2.05	2.05	2.05	2.06	2.07	2.09	2.09	2.10	2.10

T.C. = Total Heating Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.

Highlight = Tested Data

5. EXPANDED RATINGS (CONTINUED)

OUTDOOR UNIT: MODEL NO. SPHP-Z-V036BB AC PRO NO. 84233

INDOOR UNIT: MODEL NO. AIRH-Z-V036V12C AC PRO NO. 84228

COOLING MODE

Indoor Airflow (CFM)		1059									1235								
Outdoor DB (°F)	Indoor WB (°F)	60.8			64.4			67			60.8			64.4			67		
	Indoor DB (°F)	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80
77	T.C.	29.88	29.92	29.95	31.68	31.72	31.75	33.57	33.62	33.65	32.66	32.71	32.74	34.63	34.67	34.71	36.70	36.75	36.79
	S.C.	14.23	16.72	19.02	15.99	18.79	21.37	17.97	21.11	24.01	17.01	19.98	22.73	19.11	22.45	25.54	21.47	25.22	28.70
	I.P.	1.84	1.85	1.85	1.96	1.96	1.96	2.07	2.08	2.08	2.02	2.02	2.02	2.14	2.14	2.14	2.27	2.27	2.27
82	T.C.	30.09	30.13	30.17	31.91	31.94	31.98	33.81	33.86	33.89	32.90	32.94	32.98	34.88	34.92	34.95	36.96	37.01	37.05
	S.C.	14.52	17.05	19.40	16.31	19.16	21.80	18.33	21.53	24.49	17.35	20.38	23.19	19.50	22.90	26.05	21.90	25.73	29.27
	I.P.	2.06	2.06	2.07	2.19	2.19	2.19	2.32	2.32	2.32	2.25	2.26	2.26	2.39	2.39	2.40	2.53	2.54	2.54
86	T.C.	31.25	31.29	31.33	33.13	33.17	33.21	35.11	35.16	35.20	34.16	34.21	34.25	36.22	36.27	36.30	38.39	38.44	38.48
	S.C.	15.27	17.94	20.40	17.16	20.15	22.93	19.27	22.64	25.76	18.24	21.43	24.38	20.50	24.09	27.40	23.03	27.06	30.78
	I.P.	2.41	2.41	2.41	2.55	2.56	2.56	2.71	2.71	2.71	2.63	2.64	2.64	2.79	2.80	2.80	2.96	2.96	2.97
95	T.C.	33.86	33.90	33.94	35.90	35.94	35.98	38.04	38.09	38.14	37.01	37.06	37.10	39.24	39.29	39.33	41.59	41.64	41.69
	S.C.	16.54	19.43	22.11	18.59	21.84	24.84	20.88	24.53	27.91	19.77	23.22	26.42	22.21	26.10	29.68	24.95	29.32	33.35
	I.P.	3.19	3.19	3.20	3.38	3.38	3.39	3.58	3.59	3.59	3.49	3.49	3.49	3.70	3.70	3.70	3.92	3.92	3.93
104	T.C.	30.33	30.37	30.41	32.16	32.20	32.23	34.08	34.13	34.17	33.16	33.21	33.24	35.16	35.20	35.24	37.26	37.31	37.35
	S.C.	15.19	17.84	20.30	17.07	20.05	22.81	19.17	22.53	25.63	18.15	21.33	24.26	20.40	23.96	27.26	22.91	26.92	30.63
	I.P.	3.18	3.18	3.18	3.37	3.37	3.37	3.57	3.57	3.58	3.47	3.48	3.48	3.68	3.69	3.69	3.90	3.91	3.91
110	T.C.	28.22	28.26	28.29	29.92	29.95	29.98	31.71	31.75	31.78	30.85	30.89	30.92	32.71	32.75	32.78	34.66	34.71	34.75
	S.C.	14.65	17.21	19.58	16.46	19.34	21.99	18.49	21.72	24.71	17.50	20.56	23.39	19.67	23.11	26.28	22.10	25.96	29.53
	I.P.	3.17	3.17	3.18	3.36	3.36	3.37	3.56	3.57	3.57	3.46	3.47	3.47	3.67	3.68	3.68	3.89	3.90	3.90
115	T.C.	25.01	25.04	25.07	26.52	26.55	26.57	28.10	28.14	28.17	27.34	27.38	27.41	28.99	29.02	29.05	30.72	30.76	30.79
	S.C.	13.13	15.43	17.55	14.76	17.34	19.72	16.58	19.48	22.16	15.69	18.44	20.98	17.64	20.72	23.57	19.81	23.28	26.48
	I.P.	2.87	2.87	2.87	3.04	3.04	3.05	3.22	3.22	3.23	3.13	3.14	3.14	3.32	3.33	3.33	3.52	3.53	3.53
122	T.C.	20.73	20.76	20.78	21.98	22.00	22.03	23.29	23.32	23.35	22.66	22.69	22.72	24.03	24.06	24.08	25.46	25.50	25.52
	S.C.	11.14	13.09	14.89	12.52	14.71	16.73	14.06	16.52	18.79	13.31	15.64	17.79	14.96	17.57	19.99	16.81	19.74	22.46
	I.P.	2.46	2.47	2.47	2.61	2.61	2.62	2.77	2.77	2.77	2.69	2.70	2.70	2.85	2.86	2.86	3.02	3.03	3.03

T.C. = Total Cooling Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.

S.C. = Sensible Cooling Capacity (BTU/h x 1000) Highlight = Tested Data

HEATING MODE

Indoor Airflow (CFM)			1059					1235				
Outdoor DB (°F)	Outdoor WB (°F)	Indoor DB (°F)	60.8	68	70	71.6	75.2	60.8	68	70	71.6	75.2
5	3	T.C.	31.30	30.67	30.52	30.22	29.95	34.22	33.53	33.36	33.23	32.93
		I.P.	4.41	4.45	4.46	4.47	4.48	4.50	4.54	4.55	4.56	4.57
17	15	T.C.	31.00	30.38	30.23	29.93	29.66	33.89	33.21	33.05	32.91	32.62
		I.P.	4.15	4.19	4.20	4.20	4.22	4.24	4.27	4.28	4.29	4.31
35	33	T.C.	34.45	33.77	33.60	33.26	32.96	37.66	36.91	36.73	36.58	36.25
		I.P.	3.48	3.51	3.51	3.52	3.54	3.55	3.58	3.59	3.59	3.61
47	43	T.C.	35.98	35.26	35.08	34.73	34.42	39.33	38.54	38.35	38.20	37.85
		I.P.	3.01	3.04	3.04	3.05	3.06	3.07	3.10	3.11	3.11	3.12
62	56.5	T.C.	24.89	24.39	24.27	24.03	23.82	27.21	26.67	26.53	26.43	26.19
		I.P.	2.08	2.10	2.11	2.11	2.12	2.13	2.14	2.15	2.15	2.16

T.C. = Total Heating Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.

Highlight = Tested Data

5. EXPANDED RATINGS (CONTINUED)

OUTDOOR UNIT: MODEL NO. SPHP-Z-V048BB AC PRO NO. 84234
INDOOR UNIT: MODEL NO. AIRH-Z-V048V16C AC PRO NO. 84229

COOLING MODE

Indoor Airflow (CFM)		1441									1600								
Outdoor DB (°F)	Indoor WB (°F)	60.8			64.4			67			60.8			64.4			67		
	Indoor DB (°F)	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80
77	T.C.	38.60	38.65	38.69	40.93	40.98	41.02	43.37	43.43	43.48	41.61	41.66	41.71	44.11	44.17	44.21	46.75	46.81	46.86
	S.C.	18.65	21.91	24.92	20.96	24.62	28.00	23.54	27.66	31.46	21.66	25.45	28.95	24.35	28.60	32.53	27.35	32.13	36.55
	I.P.	2.11	2.12	2.12	2.24	2.24	2.25	2.37	2.38	2.38	2.28	2.28	2.28	2.42	2.42	2.42	2.56	2.56	2.57
82	T.C.	38.51	38.56	38.61	40.83	40.88	40.92	43.27	43.33	43.38	41.51	41.57	41.61	44.01	44.06	44.11	46.64	46.70	46.75
	S.C.	18.84	22.14	25.18	21.18	24.87	28.29	23.79	27.95	31.79	21.89	25.72	29.26	24.60	28.90	32.87	27.64	32.47	36.94
	I.P.	2.26	2.27	2.27	2.40	2.40	2.41	2.54	2.55	2.55	2.44	2.44	2.45	2.59	2.59	2.59	2.74	2.75	2.75
86	T.C.	39.91	39.97	40.01	42.32	42.37	42.41	44.85	44.90	44.95	43.02	43.08	43.12	45.61	45.67	45.71	48.34	48.40	48.45
	S.C.	19.77	23.23	26.43	22.22	26.11	29.69	24.96	29.33	33.36	22.97	26.99	30.70	25.82	30.33	34.50	29.00	34.07	38.76
	I.P.	2.80	2.81	2.81	2.97	2.97	2.98	3.15	3.15	3.16	3.02	3.02	3.03	3.20	3.21	3.21	3.39	3.40	3.40
95	T.C.	43.06	43.12	43.17	45.66	45.71	45.76	48.39	48.45	48.50	46.42	46.48	46.53	49.21	49.27	49.32	52.15	52.22	52.28
	S.C.	21.33	25.07	28.51	23.98	28.17	32.04	26.93	31.64	36.00	24.79	29.12	33.13	27.86	32.72	37.22	31.29	36.76	41.82
	I.P.	4.01	4.02	4.02	4.25	4.26	4.26	4.51	4.51	4.52	4.32	4.33	4.34	4.59	4.59	4.60	4.86	4.87	4.87
104	T.C.	39.60	39.66	39.70	41.99	42.04	42.08	44.50	44.56	44.61	42.69	42.74	42.79	45.26	45.31	45.36	47.96	48.03	48.08
	S.C.	20.11	23.63	26.88	22.60	26.55	30.20	25.39	29.83	33.93	23.36	27.45	31.23	26.26	30.85	35.09	29.50	34.66	39.43
	I.P.	4.08	4.09	4.09	4.33	4.33	4.34	4.59	4.59	4.60	4.40	4.41	4.41	4.67	4.67	4.68	4.95	4.95	4.96
110	T.C.	37.53	37.58	37.62	39.79	39.84	39.88	42.17	42.22	42.27	40.45	40.50	40.55	42.89	42.94	42.98	45.45	45.51	45.56
	S.C.	19.75	23.21	26.40	22.20	26.08	29.67	24.94	29.30	33.33	22.95	26.96	30.67	25.79	30.30	34.47	28.97	34.04	38.73
	I.P.	4.13	4.13	4.14	4.37	4.38	4.38	4.64	4.64	4.65	4.45	4.45	4.46	4.72	4.72	4.73	5.00	5.00	5.01
115	T.C.	33.02	33.06	33.10	35.01	35.05	35.09	37.10	37.15	37.19	35.59	35.64	35.68	37.74	37.78	37.82	39.99	40.04	40.09
	S.C.	17.59	20.66	23.50	19.76	23.22	26.41	22.20	26.08	29.67	20.43	24.00	27.31	22.96	26.97	30.68	25.79	30.30	34.47
	I.P.	3.85	3.86	3.86	4.08	4.09	4.09	4.33	4.33	4.34	4.15	4.16	4.16	4.40	4.41	4.41	4.66	4.67	4.68
122	T.C.	27.01	27.05	27.07	28.64	28.67	28.70	30.35	30.39	30.42	29.11	29.15	29.18	30.87	30.90	30.93	32.71	32.75	32.79
	S.C.	14.72	17.29	19.67	16.54	19.43	22.10	18.58	21.83	24.84	17.10	20.09	22.86	19.22	22.58	25.68	21.59	25.36	28.85
	I.P.	3.49	3.49	3.49	3.70	3.70	3.70	3.92	3.92	3.93	3.76	3.76	3.77	3.98	3.99	3.99	4.22	4.23	4.23

T.C. = Total Cooling Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.
S.C. = Sensible Cooling Capacity (BTU/h x 1000) Highlight = Tested Data

HEATING MODE

Indoor Airflow (CFM)			1441					1600				
Outdoor DB (°F)	Outdoor WB (°F)	Indoor DB (°F)	60.8	68	70	71.6	75.2	60.8	68	70	71.6	75.2
5	3	T.C.	45.98	45.06	44.84	44.66	44.26	49.56	48.57	48.33	48.13	47.70
		I.P.	6.52	6.58	6.59	6.60	6.63	6.59	6.64	6.66	6.67	6.70
17	15	T.C.	46.02	45.10	44.88	44.70	44.30	49.61	48.62	48.37	48.18	47.75
		I.P.	6.47	6.52	6.53	6.55	6.57	6.53	6.59	6.60	6.61	6.64
35	33	T.C.	48.38	47.42	47.18	46.99	46.57	52.15	51.11	50.85	50.65	50.19
		I.P.	5.99	6.04	6.05	6.06	6.09	6.05	6.10	6.11	6.12	6.15
47	43	T.C.	50.87	49.86	49.61	49.41	48.96	54.84	53.74	53.47	53.26	52.78
		I.P.	4.69	4.73	4.74	4.75	4.76	4.74	4.77	4.78	4.79	4.81
62	56.5	T.C.	44.48	43.59	43.38	43.20	42.81	47.95	46.99	46.75	46.57	46.15
		I.P.	3.13	3.16	3.16	3.17	3.18	3.16	3.19	3.19	3.20	3.21

T.C. = Total Heating Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.
Highlight = Tested Data

5. EXPANDED RATINGS (CONTINUED)

OUTDOOR UNIT: MODEL NO. SPHP-Z-V060BB AC PRO NO. 84235
INDOOR UNIT: MODEL NO. AIRH-Z-V060V18C AC PRO NO. 84230

COOLING MODE

Indoor Airflow (CFM)		1647									1800								
Outdoor DB (°F)	Indoor WB (°F)	60.8			64.4			67			60.8			64.4			67		
	Indoor DB (°F)	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80	73.4	77	80
77	T.C.	47.25	47.31	47.36	50.09	50.15	50.20	53.09	53.15	53.21	50.13	50.20	50.25	53.15	53.21	53.27	56.33	56.40	56.46
	S.C.	23.18	27.24	30.99	26.05	30.61	34.81	29.27	34.39	39.12	26.10	30.67	34.88	29.33	34.46	39.20	32.95	38.71	44.04
	I.P.	2.88	2.89	2.89	3.06	3.06	3.06	3.24	3.24	3.25	3.06	3.06	3.07	3.24	3.25	3.25	3.44	3.44	3.44
82	T.C.	48.11	48.18	48.23	51.01	51.07	51.12	54.06	54.13	54.19	51.05	51.12	51.17	54.13	54.19	54.24	57.36	57.44	57.50
	S.C.	23.91	28.09	31.96	26.87	31.57	35.91	30.19	35.47	40.35	26.92	31.63	35.98	30.26	35.54	40.43	33.99	39.93	45.42
	I.P.	3.22	3.22	3.22	3.41	3.41	3.42	3.61	3.62	3.62	3.41	3.42	3.42	3.62	3.62	3.63	3.83	3.84	3.84
86	T.C.	49.14	49.20	49.26	52.10	52.16	52.21	55.21	55.28	55.34	52.14	52.21	52.26	55.28	55.34	55.40	58.58	58.66	58.72
	S.C.	24.73	29.05	33.05	27.79	32.65	37.14	31.22	36.68	41.73	27.84	32.71	37.21	31.29	36.76	41.81	35.15	41.29	46.98
	I.P.	3.96	3.96	3.97	4.20	4.20	4.21	4.45	4.45	4.46	4.20	4.20	4.21	4.45	4.46	4.46	4.72	4.72	4.73
95	T.C.	51.44	51.51	51.56	54.54	54.60	54.66	57.80	57.87	57.94	54.58	54.65	54.71	57.87	57.94	57.99	61.33	61.41	61.47
	S.C.	25.89	30.42	34.60	29.09	34.18	38.88	32.68	38.40	43.68	29.15	34.24	38.95	32.76	38.48	43.77	36.80	43.23	49.18
	I.P.	5.63	5.63	5.64	5.96	5.97	5.98	6.32	6.33	6.34	5.97	5.98	5.98	6.33	6.34	6.34	6.71	6.72	6.72
104	T.C.	47.07	47.14	47.19	49.91	49.97	50.02	52.89	52.96	53.02	49.95	50.01	50.07	52.96	53.02	53.07	56.12	56.19	56.26
	S.C.	24.28	28.53	32.45	27.29	32.06	36.47	30.66	36.02	40.97	27.34	32.12	36.54	30.72	36.09	41.06	34.51	40.55	46.13
	I.P.	5.55	5.56	5.56	5.88	5.89	5.89	6.23	6.24	6.25	5.89	5.89	5.90	6.24	6.25	6.25	6.61	6.62	6.63
110	T.C.	44.45	44.51	44.56	47.13	47.19	47.23	49.95	50.01	50.07	47.17	47.23	47.28	50.01	50.07	50.12	53.00	53.07	53.12
	S.C.	23.77	27.93	31.77	26.71	31.38	35.70	30.01	35.26	40.11	26.76	31.44	35.77	30.08	35.33	40.19	33.79	39.69	45.16
	I.P.	5.50	5.51	5.51	5.83	5.84	5.84	6.18	6.19	6.20	5.84	5.84	5.85	6.19	6.20	6.20	6.56	6.57	6.57
115	T.C.	37.50	37.55	37.59	39.76	39.81	39.85	42.14	42.19	42.24	39.79	39.84	39.89	42.19	42.24	42.28	44.71	44.77	44.82
	S.C.	20.29	23.84	27.12	22.80	26.78	30.47	25.61	30.09	34.23	22.84	26.84	30.53	25.67	30.16	34.30	28.84	33.88	38.54
	I.P.	4.63	4.63	4.64	4.91	4.91	4.92	5.20	5.21	5.21	4.91	4.92	4.92	5.21	5.21	5.22	5.52	5.53	5.53
122	T.C.	28.23	28.27	28.30	29.93	29.97	30.00	31.72	31.76	31.80	29.95	29.99	30.03	31.76	31.80	31.83	33.66	33.70	33.74
	S.C.	15.63	18.36	20.89	17.56	20.63	23.47	19.73	23.18	26.37	17.59	20.67	23.52	19.77	23.23	26.42	22.21	26.10	29.69
	I.P.	3.46	3.47	3.47	3.67	3.68	3.68	3.89	3.90	3.90	3.68	3.68	3.69	3.90	3.90	3.91	4.13	4.14	4.14

T.C. = Total Cooling Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.
S.C. = Sensible Cooling Capacity (BTU/h x 1000) Highlight = Tested Data

HEATING MODE

Indoor Airflow (CFM)			1647					1800				
Outdoor DB (°F)	Outdoor WB (°F)	Indoor DB (°F)	60.8	68	70	71.6	75.2	60.8	68	70	71.6	75.2
5	3	T.C.	54.93	53.83	53.56	53.35	52.87	58.28	57.12	56.83	56.60	56.09
		I.P.	7.86	7.92	7.94	7.95	7.98	7.94	8.00	8.02	8.03	8.06
17	15	T.C.	52.46	51.41	51.16	50.95	50.49	55.67	54.55	54.28	54.06	53.58
		I.P.	6.79	6.84	6.85	6.87	6.90	6.85	6.91	6.92	6.94	6.97
35	33	T.C.	46.01	45.09	44.86	44.48	44.07	51.80	50.76	50.51	50.31	49.85
		I.P.	5.78	5.83	5.84	5.85	5.88	5.84	5.89	5.90	5.91	5.94
47	43	T.C.	48.82	47.84	47.60	47.41	46.98	61.17	59.95	59.65	59.41	58.88
		I.P.	5.61	5.65	5.66	5.68	5.70	5.66	5.71	5.72	5.73	5.76
62	56.5	T.C.	50.02	49.02	48.78	48.58	48.14	53.08	52.02	51.76	51.55	51.08
		I.P.	4.19	4.22	4.23	4.24	4.26	4.23	4.27	4.27	4.28	4.30

T.C. = Total Heating Capacity (BTU/h x 1000) I.P. = Electrical Input Power (kW). Includes indoor and outdoor units.
Highlight = Tested Data